

Faculty: BSC(Hons) Computer System Engineering

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MODULE CODE: CET138

MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

SUBMISSION DATE AND TIME: Fri, 19 Jun 2026

ASSIGNMENT: 1

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CET138 Full Stack Development

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Live Demo Link of CSS:

<https://rahulchaudhary12-debug.github.io/Demos/demo-css.html>

What is CSS ?

HTML's appearance is determined by CSS (Cascading Style Sheets). The Code Craft CSS demo (demo-css.html) is an advanced demonstration of CSS3 techniques that is 100% CSS and has no JavaScript. It shows 8 different visual effects: animated gradient hero, typing text, hover cards, skill progress bars, floating shapes, gradient buttons, glass morphism and CSS-only loaders.

What are CSS variables and how to set them up? What is CSS variable and how to set base setup?

Consistent color palette is configured using CSS custom properties (variables) for all eight sections:

```
/* Defined in style.css - used across codecraft.html */
:root {
  --accent: #3af0c0; /* teal - primary highlight */
  --accent2: #8b5cf6; /* purple */
  --accent3: #ff4f87; /* pink */
  --gold: #f0c940; /* gold */
  --bg: #0a0a0f; /* near-black background */
  --surface: #12121a; /* card background */
  --text: #f0f0f5; /* off-white text */
  --muted: rgba(240,240,240,0.45); /* subdued text */
  --radius: 16px;
}

*, *::before, *::after { box-sizing: border-box; margin:0; padding:0; }
body { background: var(--bg); color: var(--text); font-family: 'DM Sans', system-ui; }
```

Key CSS Techniques Demonstrated

Animated Gradient Hero (@keyframes + animation)

```
@keyframes gradientShift {
  0% { background-position: 0% 50%; }
  50% { background-position: 100% 50%; }
  100% { background-position: 0% 50%; }
}
@keyframes fadeSlideUp {
  from { opacity: 0; transform: translateY(28px); }
  to { opacity: 1; transform: translateY(0); }
}
.hero-banner {
  background: linear-gradient(270deg, #3af0c0, #8b5cf6, #ff4f87, #f0c940, #3af0c0);
  background-size: 400% 400%;
  animation: gradientShift 8s ease infinite;
}
.hero-content { animation: fadeSlideUp 0.8s ease both; }
```

CSS Typing Effect (no JavaScript)

```
@keyframes typing { from { width:0 } to { width:100% } }
@keyframes blink { 0%,100%{opacity:1} 50%{opacity:0} }

.typing-text {
  overflow: hidden;
  white-space: nowrap;
  border-right: 3px solid #3af0c0; /* cursor */
  width: 0;
  animation: typing 2.4s steps(30) 0.4s forwards,
            blink 0.7s step-end infinite;
}
```

Hover Cards (transform + radial-gradient :before)

```
.hcard {
  transition: transform .3s ease, box-shadow .3s ease, border-color .3s ease;
  position: relative; overflow: hidden;
}
.hcard::before { /* invisible overlay revealed on hover */
  content: '';
  position: absolute; inset: 0;
  opacity: 0;
  transition: opacity .3s ease;
}
.hcard-1::before { background: radial-gradient(circle at top, rgba(58,240,192,.12),
transparent 70%); }
.hcard-1:hover { transform: translateY(-8px); border-color: rgba(58,240,192,.35); }
.hcard-1:hover::before { opacity: 1; }
```

Animated Progress + Floating Shapes

```
/* Animated progress bars */  
  
@keyframes barFill { from { width: 0; } }  
  
.bar-fill.c1 { width:92%; background:linear-  
gradient(90deg,#3af0c0,#8b5cf6); animation: barFill 1.4s cubic-  
bezier(.4,0,.2,1) both; }  
  
/* Floating + pulsing shapes */  
  
@keyframes float { 0%,100% { transform: translateY(0); } 50% {  
transform: translateY(-14px); } }  
  
@keyframes pulse { 0%,100% { box-shadow: 0 0 0 0  
rgba(58,240,192,.45); } 50% { box-shadow: 0 0 0 18px  
rgba(58,240,192,0); } }  
  
@keyframes spin { to { transform: rotate(360deg); } }  
  
.shape-circle { animation: float 3s ease-in-out infinite, pulse 3s ease-  
in-out infinite; }  
  
.shape-ring { animation: spin 4s linear infinite; }
```

Glass morphism Effect

```
.glass-card {  
  background: rgba(255,255,255,0.06);  
  border: 1px solid rgba(255,255,255,0.14);  
  backdrop-filter: blur(16px); /* frosted glass blur */  
  border-radius: 18px;  
  padding: 32px;  
  transition: transform .3s;  
}
```

```
.glass-card:hover { transform: translateY(-4px); }  
/* Three colour orbs behind the card create the coloured glow */  
.glass-orb { position: absolute; border-radius: 50%; filter: blur(60px); }
```

Summary

The CodeCraft CSS demo showcases that CSS alone, without any JavaScript, can be used to create rich, interactive and animated user interfaces. It supports most of the CSS3 features, such as:

- Custom properties in CSS for a consistent and maintainable design system
- The gradientShift, fadeSlideUp, typing, barFill, float, pulse, spin, and ripple are all in-built @keyframes animations.
- The following are in-built @keyframes animations: gradientShift, fadeSlideUp, typing, barFill, float, pulse, spin, ripple

Can use pseudo-elements (::before, ::after) for hover overlays and decorative content.

- Hover lifts and spin effects using CSS transforms: translateY, scale, rotate
- It combines backdrop-filter: blur with semi-transparent backgrounds, which is known as Glass morphism.

- These cards are designed with CSS Grid and Flexbox. Card grid, shapes area, loaders row and swatches are designed with CSS Grid and Flexbox.
- The clamp() function is used in fluid typography where it scales between the viewports.